



MAARULA CLASSES

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NIMCET PAPER

G.P.

PYQ



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1. Find the value of 'x', if $\left(\frac{1}{2^{\log_x 4}}\right)\left(\frac{1}{2^{\log_x 16}}\right)\left(\frac{1}{2^{\log_x 256}}\right) \dots \infty = 2$ (GP)

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- (a)2 (b)1/2 (c)4 (d)1/4

Solution:

1. (c) From given relation, we get

$$\text{Then, } 2^{\log_x 4 + \log_x 16 + \log_x 256 + \dots \infty} = \frac{1}{2}$$

$$\Rightarrow 2^{\frac{1}{\log_x}(\log 4 + \log 4^2 + \log 4^4 + \dots \infty)} = \frac{1}{2}$$

$$\Rightarrow 2^{\frac{\log 4}{\log x}(1 + 2 + 4 + \dots)} = \frac{1}{2}$$

$$\Rightarrow 2^{\frac{\log 4}{\log x} \left\{ \frac{1}{1-2} \right\}} = \frac{1}{2}$$

$$\Rightarrow 2^{-\log x 4} = 2^{-1}$$

$$\Rightarrow \log_x 4 = 1$$

$$\Rightarrow x = 4$$

